

THE UNIVERSITY *of* EDINBURGH

Week 1: Understanding the Need for Machine Learning

Foundations of Machine Learning
for Health and Social Care Tutorial

Audio Check

Can you hear the presenter talking?

Please type yes or no in the “Text chat area” or react in some way.

If you can’t hear:

- Check your audio/visual settings in the collaborate panel.
- Try signing out and signing back into the session.
- Type into the chat box and a member of the teaching team will try to assist you.

Asking Questions

Please always use the hand raise icon to queue up your question if you would like to ask a question verbally.

Alternatively you can put a question into the chat at any point in time.

Recording

The session will now be recorded. Any further information that you provide during the session is optional and in doing so you give us consent to process this information.

These sessions will be stored by The University of Edinburgh for one year and published for 30 days after the event. Schools or Services may use the recordings for up to a year on relevant websites.

By taking part in a session, you give us your consent to utilise any information you provide during the session.



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Course Overview / Admin

Course Structure

Five weeks of content.
Released a week in advance.
Fridays at 1400 UTC/GMT.
Quizzes all weeks that are formative.
Some further optional readings and resources.

Blackboard®
LEARN ULTRA

Foundations of Machine Learning for Health and Social Care

HEIN11098 Course Notes

Mr Christopher A Oldnall



Edinburgh Medical School
The University of Edinburgh
2026 Delivery

Course Structure

Week	Week commencing	Topic	Collaborate sessions
1	02/11/2026	Understanding the need for machine learning	04/11/2026 17:00 [JW]
2	09/11/2026	Quantitative-based problems	11/11/2026 17:00 [JW]
3	16/11/2026	Qualitative-based problems	18/11/2026 17:00 [JW]
		Drop-in Session	20/11/2026 12:00 [JW]
4	23/11/2026	Unsupervised learning	25/11/2026 17:00 [JW]
5	30/11/2026	Deep learning	02/12/2026 17:00 [JW]
		Drop-In	07/12/2026 12:00 [JW]
	14/12/2026	DEADLINE: Individual project report	14/12/2026

Live Sessions and Support

Five live tutorials on Wednesdays at 17:00-18:30 GMT/UTC.

Two drop-in sessions to support with learning and/or assessment on

- Friday 20th November at 12:00
- Monday 7th December at 12:00

Blackboard collaborate™

Who might you interact with?

Course Lead: John Wilson



Course Secretary: Sabrina (Yingya) Xu
(UsherTO.Admin@ed.ac.uk)



Course Organiser: Brittany Blankinship



Assessment for this module

One programming report.
Released start of Week 4 (23/11/2026).
Due three weeks later (14/12/2026).

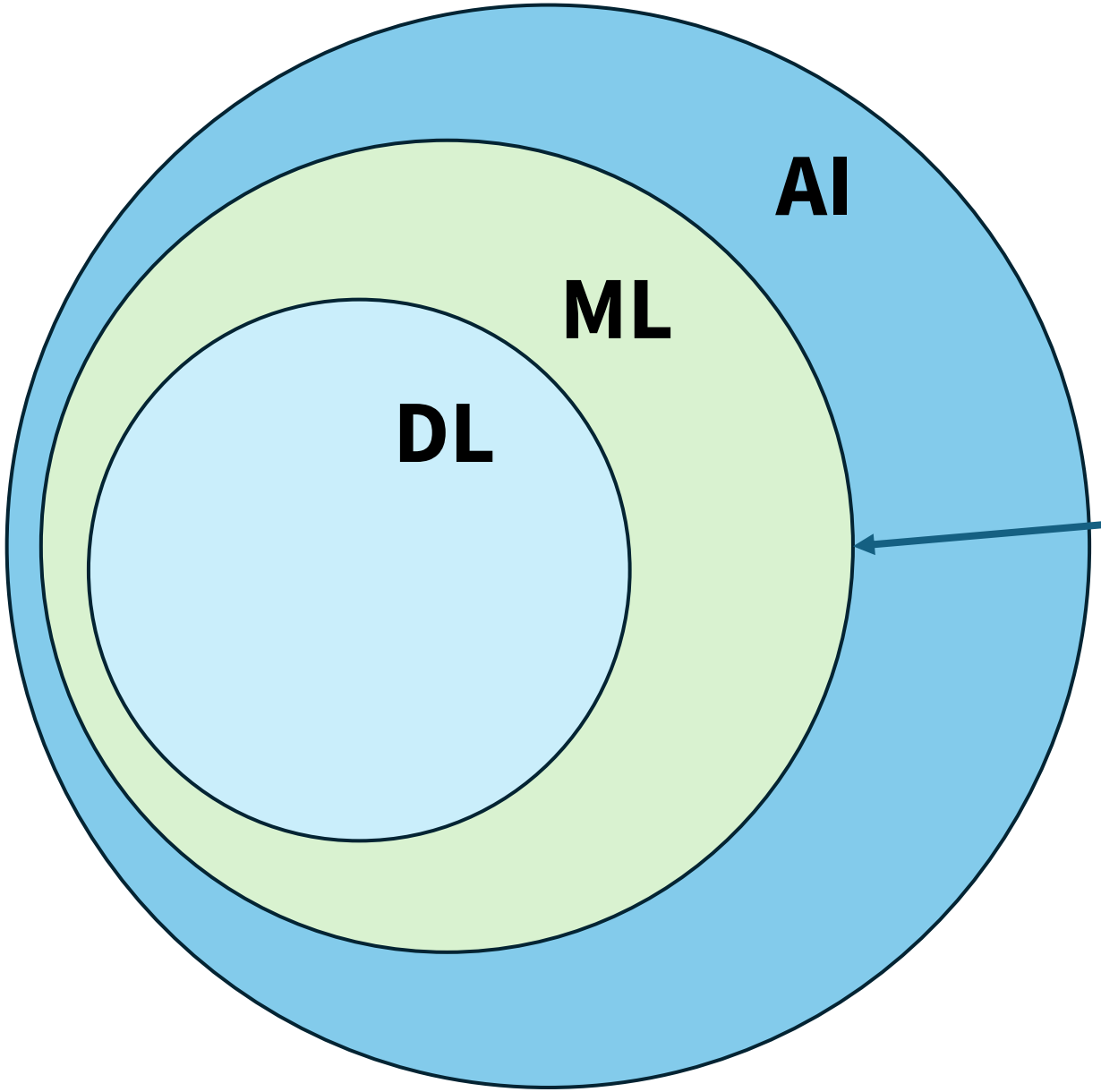
The final programming report will assess all learning outcomes for this course. It will have two parts:

Part A: Case Study Ethics Evaluation (max 1000 word essay)
Part B: Machine Learning in Python Notebook (4 tasks)



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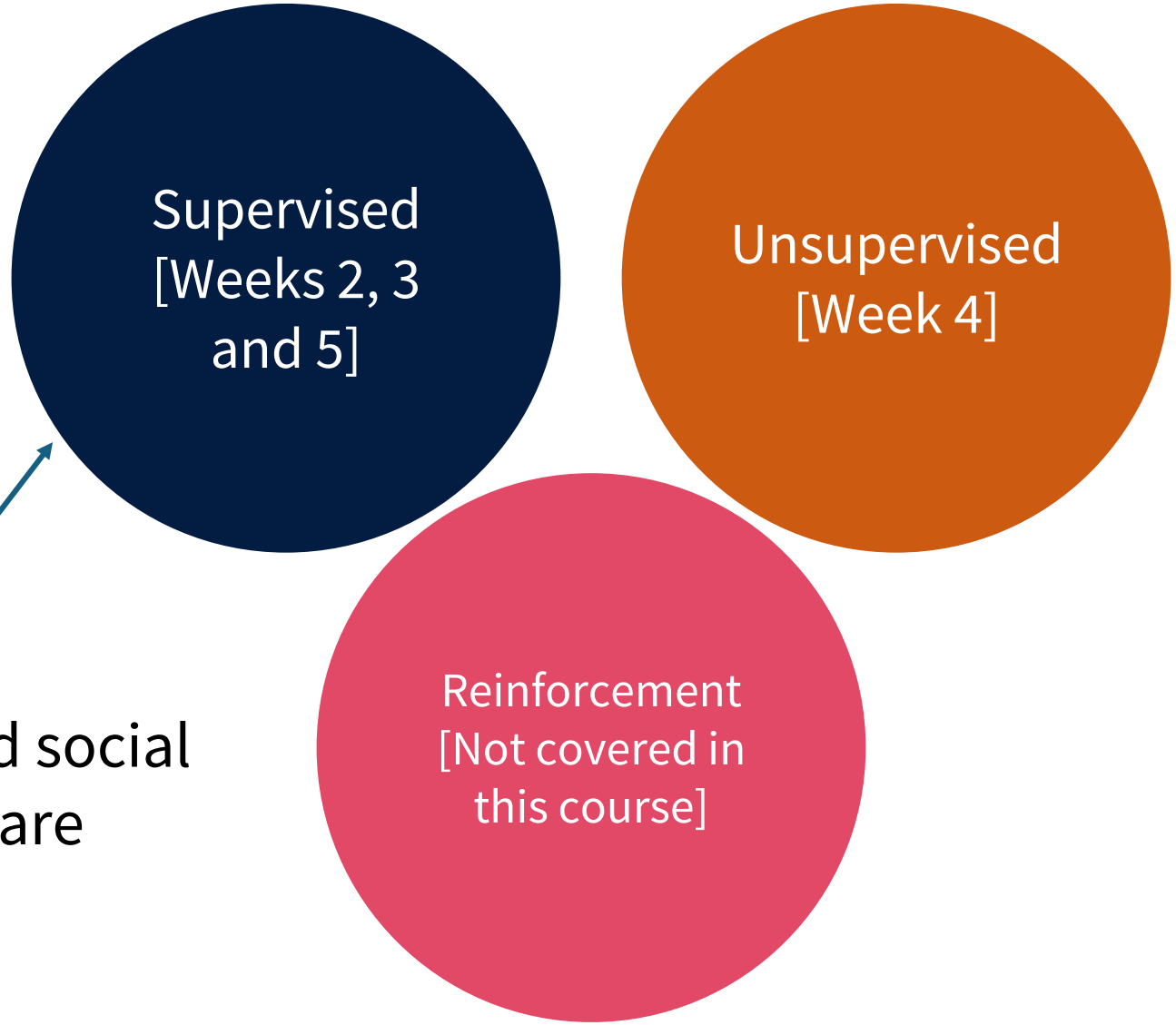
Week One: Understanding the Need for Machine Learning



How do AI – ML – DL connect?

Machine learning is the branch of AI which replaces rule-writing with data-driven inference.

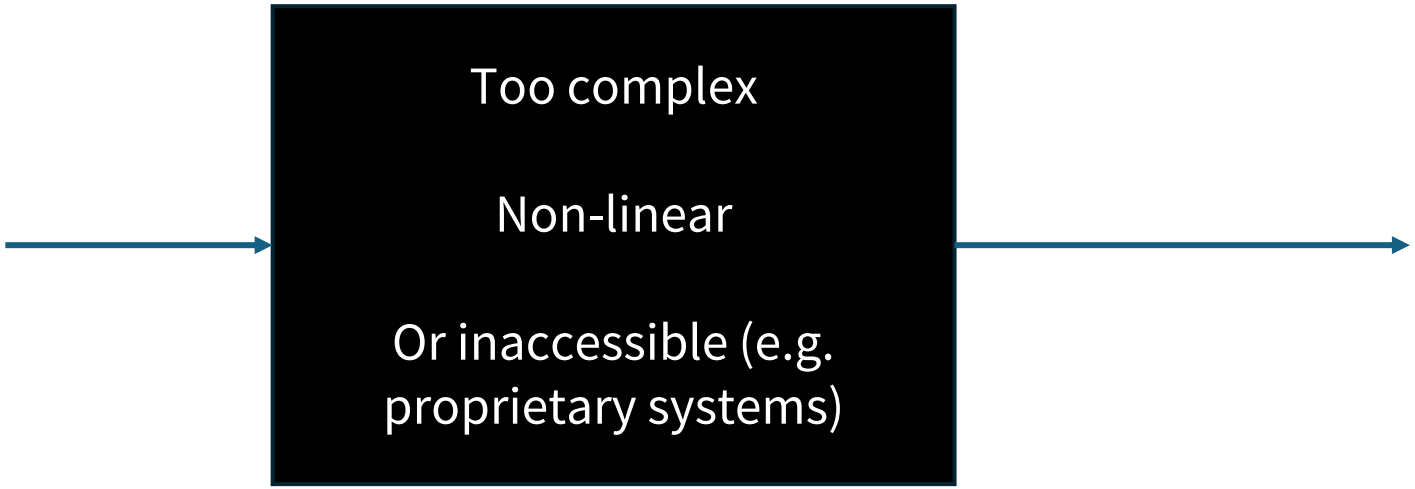
Types of ML



Most of health and social care applications are supervised ML.

The black box problem.

A black box model produces predictions without transparent reasoning



Group-Level Algorithmic Bias

Let $S \in \{0,1\}$ be a 'sensitive attribute' (for example gender or ethnicity)

A model is biased if:

$$E\left[\left|\hat{f}(\mathbf{x}) - y\right| \mid S = 0\right] \neq E\left[\left|\hat{f}(\mathbf{x}) - y\right| \mid S = 1\right]$$

That is the errors themselves are unevenly distributed between the different groups.

So what is algorithmic confounding?

Models trained on observational data may:

- Exploit correlations
- Learn spurious associations
- Mistake correlation for causation

Predictions can look impressive…
and still be wrong.

Why ML has an environmental cost

Modern ML increasingly relies on:

- Large ensembles
- Deep neural networks
- Extensive hyperparameter tuning

Consequences:

- High energy consumption
- Significant carbon emissions
- Repeated training and retraining

Global vs local explanability

Two complementary questions:

Global explanations:
What patterns does the model learn overall?

Local explanations:
Why was this specific prediction made?

What is generative AI?

Learn the distribution of data
Generate new samples from that distribution

They can produce:

- Text
- Images
- Audio
- Synthetic records

Key distinction:
Discriminative models decide.
Generative models create.

Generative AI:

- Amplifies ethical risks
- Challenges accountability
- Blurs evidence and invention

Just because we can generate data, does not mean we should deploy it without restraint.

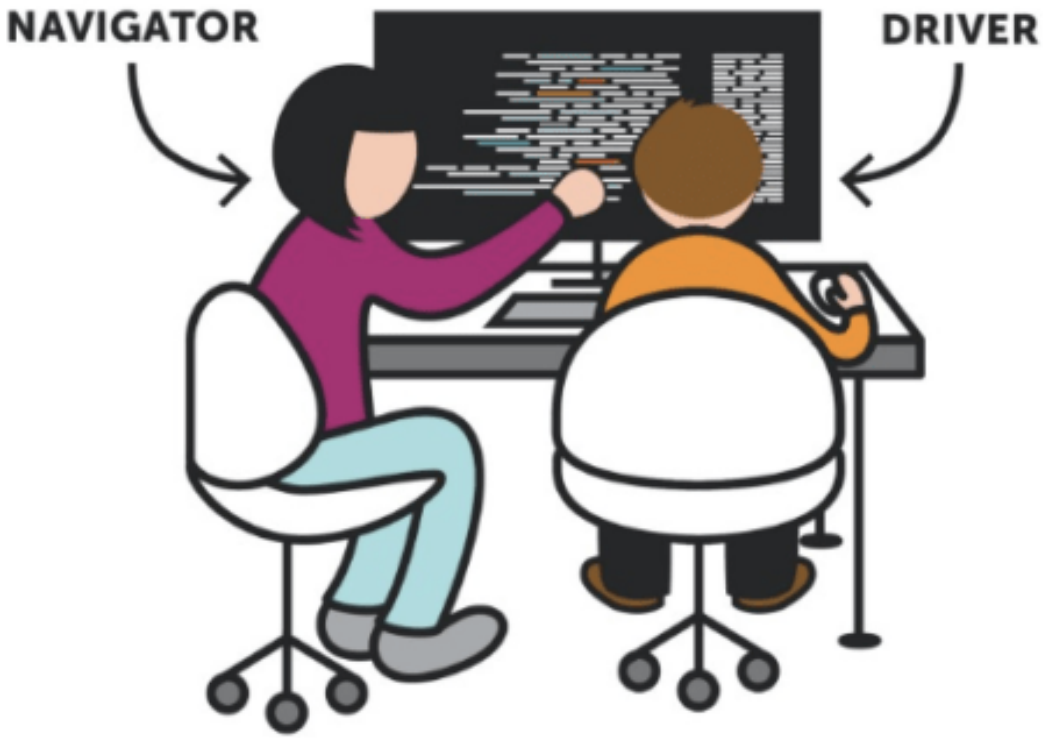
Any Questions?



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Paired Programming Practice

PAIR PROGRAMMING











- Identify ethical risks associated with machine learning and generative AI in health and social care.
- Distinguish prediction from explanation and causation, and recognise why this matters for trust and accountability.
- Use simple Python operations to explore how technical choices can influence fairness, transparency, and interpretation.